



Fix the Code

Find the wrong block and fix it

Name:

Date:

★ I can find the wrong block in the code and fix it.

This code has a bug

This code is supposed to land Bit on 4, but one block is **WRONG**, so Bit misses it. A coder runs the code, finds the wrong block, and fixes it. Bit starts on 1. The target is 4.

Bit's number path (target is 4)



Buggy code — should reach 4

start at 1

forward 2

back 1

1 Run it

Run the buggy code from 1: forward 2, then back 1. What number does Bit land on?
Is it the target (4)?

2 Find the wrong block

One block moves Bit the WRONG way and sends it away from 4. Circle the block that is wrong.

Fixed code

start at 1

forward 2



Write the block that makes Bit reach 4.

3 You try — fix it

Write the correct block in the empty space so Bit reaches 4. Then run the fixed code. What number does Bit land on now?

Big idea

Finding and fixing a wrong block is called debugging. Run the code, see Bit miss the target, find the block that is wrong, fix it, and run it again.



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Quick facilitation notes & answer key

What this worksheet teaches

Students debug a set of instructions: run it, see Bit miss the target, find the one wrong block, fix it, and run again to check. Run it, find the wrong block, fix it, run again — a first debugging routine. No coding experience needed.

Before you start (no computer needed)

Paper and pencil only. Optional: a floor number line 0 to 5 with a marker on 4 (the target) so students can walk the buggy code and feel Bit go the wrong way.

How to lead it (about 20 minutes)

1. Show them first (you do). Run the buggy code: "start 1... forward 2 to 3... back 1 to 2. Bit is on 2, but the target is 4 — something is wrong."
2. Do one together (we do). Exercise 1: the buggy code lands on 2 (not 4). Exercise 2: the back 1 block moves Bit away from 4 — circle it.
3. Let them try alone (you do). Exercise 3: replace the wrong block with forward 1; the code becomes start 1, forward 2, forward 1 and reaches 4.
4. Big idea: "Run it, find the wrong block, fix it, run again — that is how coders debug."

Answer key (these are the verified correct answers)

Exercise 1 — buggy code: 1 → forward 2 → 3 → back 1 → 2. Bit lands on 2, not the target 4.

Exercise 2 — the wrong block is back 1: it sends Bit backward, away from 4.

Exercise 3 — fix it to forward 1: 1 → 3 → 4. Bit lands on the target, 4.

Why this is the bug: the first two blocks bring Bit to 3, one short of 4; a forward 1 finishes the job, but a back 1 undoes progress and lands on 2.

If students get stuck — and ways to adjust

Watch for: blaming the start or the forward 2. Have students run step by step and watch exactly where Bit turns away from 4.

Make it easier: walk the buggy code on the floor line, then swap the last card to forward 1 and re-walk.

Make it harder: ask for a DIFFERENT single-block fix that also reaches 4 — e.g. change forward 2 to forward 4 and keep back 1: 1 → 5 → 4.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions someone wrote, change one, and explain what the change did.

You'll know it worked when

The student lands the buggy code on 2, identifies back 1 as the wrong block, and fixes it to forward 1 to reach 4.